a possibility of selecting the same sample any number of times. So, simple random sampling without replacement is followed.

Thus in simple random sampling from a population of N units, the probability of drawing any unit at the first draw is $\frac{1}{N}$, the probability of drawing any unit in the second draw from among the available $(N-1)$ units is $\frac{1}{(N-1)}$, and so on. . Several methods have been adopted for random selection of the samples from the population. Of those, the following two methods are generally used and which are described below.

(A) Lottery method

This is the most popular and simplest method when the population is finite. In this method, all the items of the population are numbered on separate slips of paper of same size, shape and colour. They are folded and placed in a container and shuffled thoroughly. Then the required numbers of slips are selected for the desired sample size. The selection of items thus depends on chance.

For example, if we want to select 10 students, out of 100 students, then we must write the names/roll number of all the 100 students on slips of the same size and mix them, then we make a blindfold selection of 10 students. This method is called unrestricted random sampling, because units are selected from the population without any restriction. This method is mostly used in lottery draws. If the population or universe is infinite, this method is inapplicable.

(B) Table of Random number

When the population size is large, it is difficult to number all the items on separate slips of paper of same size, shape and colour. The alternative method is that of using the table of random numbers. The most practical, easy and inexpensive method of selecting a random sample can be done through "Random Number Table". The random number table has been so constructed that each of the digits 0,1,2,…,9 will appear approximately with the same frequency and independently of each other.

The various random number tables available are

a. L.H.C. Tippett random number series
b. Fisher and Yates random number series
c. Kendall and Smith random number series
d. Rand Corporation random number series.

Tippett's table of random numbers is most popularly used in practice. Given below the first forty sets from Tippett's table as an illustration of the general appearance of random numbers:

2952	6641	3992	9792	7969	5911	3170	5624
4167	9524	1545	1396	7203	5356	1300	2693
2670	7483	3408	2762	3563	1089	6913	7991
0560	5246	1112	6107	6008	8125	4233	8776
2754	9143	1405	9025	7002	6111	8816	6446

Suppose, if we want to select the required number of samples from a population of size N(<99) then any two digit random number can be selected from (00 to 99) from the above random number table. Similarly if N(< 999) or (< 9999), then any three digit random number or four digit random number can be selected from (000 to 999) or (0000 to 9999).

The procedure of selecting the random samples consists of following steps:

1. Identify the N units in the population with the numbers from 1 to N.
2. Select at random, any page from the 'Random Number Table'.
3. Select the required number of samples from any row or column or diagonal.

XII Std - Business Maths & Stat EM Chapter 8.indd 177 2/29/2020 11:40:02 AM